

ALPACA AI TRADING AGENTS HACKATHON — OPTIONS ALPHA

AllPath

trading agent

One kickoff conversation. Then it researches, trades stocks & options through Alpaca's MCP server — and *rewrites its own strategies every night.*

`github.com/dukesky/allpath-trading-agent` · `trading.all-path.com` · self-hosted · open source · MIT · 2,368 tests

THE PROBLEM

Analysis bots can't trade. Trading bots can't *think*.

LLM TRADING PROJECTS

Stop at "here's my analysis"

Great research, zero execution. No broker, no risk model, no accountability for whether the idea actually made money.

ALGORITHMIC FRAMEWORKS

Execute but can't reason

Fast and deterministic — but they can't read the news, question a thesis, remember what went wrong last month, or rewrite themselves.

ALLPATH

Closes the loop

An agent that researches like an analyst, executes like a machine, journals every decision, and evolves its own strategies nightly — inside guards no model can bypass.

Three loops, one process

ON DEMAND

Conversation

Web, Telegram, or CLI. The agent interviews you, researches live data, and drafts strategies as readable YAML — thesis in prose, rules as code. Anything touching money becomes an approval card.

EVERY 30 MIN · ZERO LLM COST

Sentinel

Plain code evaluates every rule against live quotes. Hard rules execute with no model in the path; soft rules wake a cheap review agent to research before deciding.

NIGHTLY · AFTER CLOSE

Reflection

A bounded agent session reviews the day against every thesis, writes lessons to curated memory, re-arms spent triggers, and revises its own strategies — as guarded diffs.

rule fires → contract selected via Alpaca MCP → risk gate (deterministic) → order → journal → push receipt → nightly reflection

Every position, every execution, *one glance*

Stocks and option legs together — OCC contracts with live value and P&L, next to the equity curve

Strategy cards — authorization tier, progress toward target weight, profit/stop chips, ”rule triggered” state

Sentinel heartbeat — last check, interval, market-open badge; a halt banner if the breaker ever trips

Recent executions & queue — what just filled, what’s waiting, straight from the journal

allpath trade

Dashboard

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PAPER

SHADOW

AMD	4.242854113	\$471.38	\$1,975.39	\$-24.60
AMD260902C00485000	2	\$5.15	\$518.00	\$-512.00
CRM260904P00252500	5	\$2.79	\$1,800.00	\$405.00
MU	2.717439509	\$919.98	\$2,534.99	\$35.00
MU260902C00965000	1	\$10.55	\$765.00	\$-290.00
NVDA	13.247692647	\$226.45	\$2,882.04	\$-117.95
NVDA260902C00230000	13	\$1.08	\$442.00	\$-962.00
TSM	4.681853318	\$427.18	\$1,954.77	\$-45.22
TSM260904C00435000	4	\$3.05	\$560.00	\$-660.00

STRATEGIES

AMD

AMD short-term swing (secondary semiconductor exposure)

\$465.58 -2.33%

active

auto 2.0% of 4.0% target

target > 505

stop < 455

rule triggered

CRM

CRM post-earnings spike fade (put hedge, software short leg of semis/software pair)

\$256.00 +1.57%

active

auto - of 3.0% target

rule triggered

Share an idea. Get a *plan*.

It researches before it agrees — live quotes, history, news; it validated our theses and rejected one with evidence

Plans become strategy drafts — YAML proposals with rationale, old-vs-new conditions, checks against live prices

Honest about boundaries — it states what it didn’t touch and why, unprompted

Same conversation everywhere — web and Telegram share one thread, one memory; screenshots can be attached and read

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✓ Four revisions drafted (#17-#20), pending your approval

Fix applied to each

Strategy	Old entry-call / entry-put gate	New gate	Ceiling checked vs live quote
nvda-momentum-swing	<code>position_weight == 0 and price < 232</code>	<code>position_weight > 0 and position_weight < 0.06 and price < 232</code>	\$223.25 — not stale, kept \$232
mu-oversold-swing	<code>position_weight == 0 and price < 950</code>	<code>position_weight > 0 and position_weight < 0.06 and price < 950</code>	\$931.53 — not stale, kept \$950
amd-swing	<code>position_weight == 0 and price < 485</code>	<code>position_weight > 0 and position_weight < 0.05 and price < 485</code>	\$472.72 — not stale, kept \$485
tsm-swing-contest	<code>position_weight == 0 and price < 435</code>	<code>position_weight > 0 and position_weight < 0.05 and price < 435</code>	\$423.66 — not stale, kept \$435

All four current live quotes sit comfortably under their original ceilings, so I re-anchored nothing — the original price gates weren't stale, only the `position_weight == 0` clause was broken by the stock fill. Exits (take-profit/stop-loss, options mirrors), budgets (\$1500 each), and `dte>=5/otm` parameters are all unchanged.

crm-fade-swing — left untouched, as you said; its `position_weight == 0 and price > 245` gate is still valid since no stock leg exists on that one.

What I didn't touch

- I didn't add any account-level drawdown logic here — as noted when these were first drafted, the 15% circuit breaker isn't expressible in a per-ticker strategy file and has to be a platform/risk-gate control.
- I kept each `max_weight` as the ceiling for the revised condition rather than inventing a new number, since that's the existing risk boundary you already approved for each name.

Please review and approve #17-#20 in Pending, then re-arm the four entry-call rules (and re-arm crm-fade-swing's entry-put) as you said you would. Let me know once they're armed and I'll check fills.

You resolved #20. Result: revision applied to tsm-swing-contest

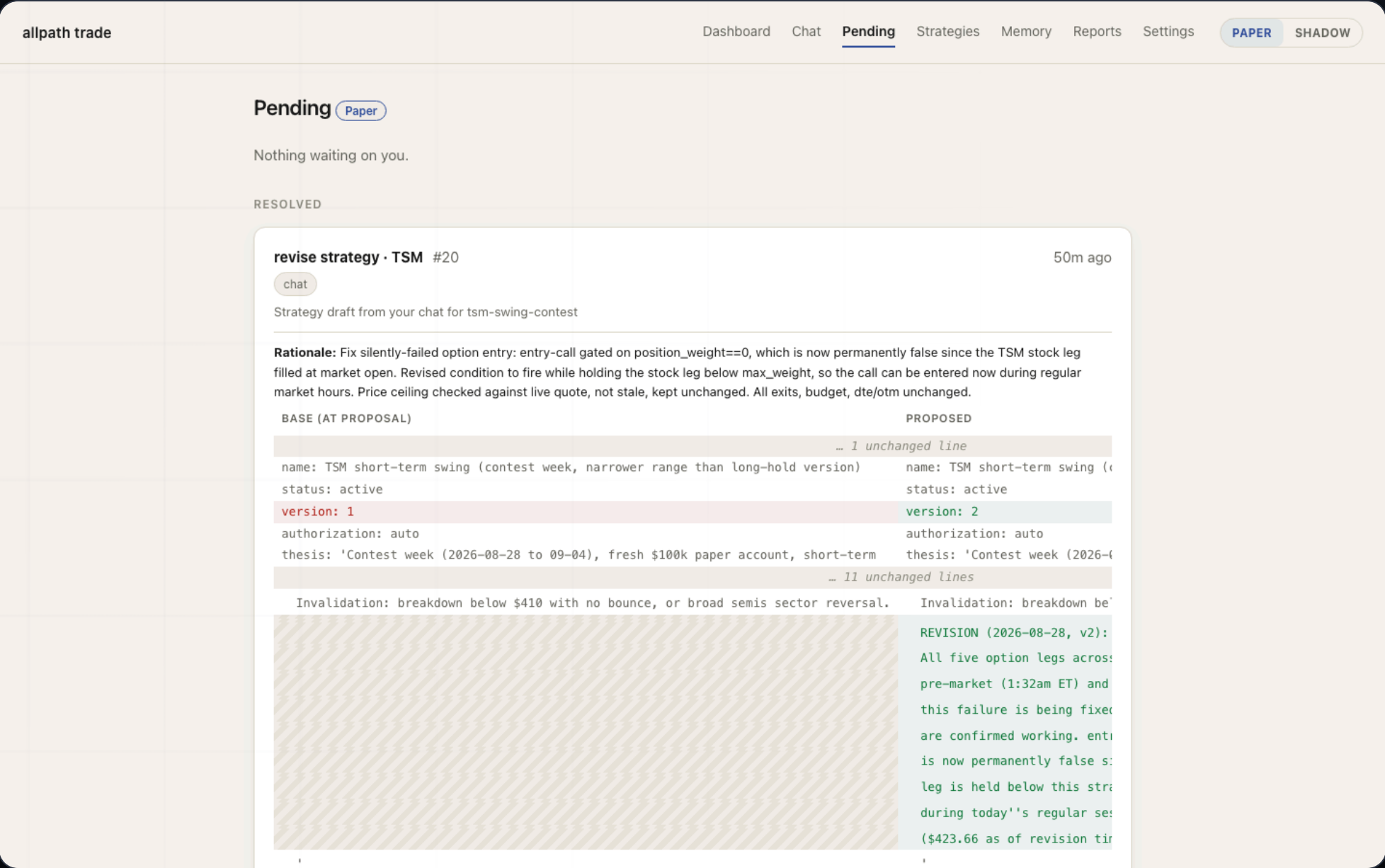
Approve from anywhere — *or not at all*

auto — this week’s mode: hard rules execute alone, straight through the risk gate; you get the receipt

confirm — proposals queue on Pending with the review agent’s analysis attached, human decides

Approve from your pocket — web card, Telegram inline buttons, or a tokenized link right in the push notification

Every change is a diff — strategy edits show base-vs-proposed side by side; stale versions are refused



The LLM writes *intent*. Code picks the contract.

01	Agent writes a rule in its strategy	<code>buy_call \$1500 dte>=5 otm=3%</code>
02	Sentinel trigger → option chain via MCP	<code>get_option_contracts · expirations ≥ today+5, calls only</code>
03	Deterministic contract selection	<code>nearest expiry · strike closest to spot × 1.03 · qty = [\$1500 / (ask × 100)]</code>
04	Risk gate — plain code, non-bypassable	<code>premium ≤ per-order cap · options exposure ≤ 10% equity · cash reserve · daily cap</code>
05	Order placed through MCP, journaled, notified	<code>place_option_order · buy_to_open · market · day</code>

- `managed MCP subprocess · respawn + timeouts`
- `DTE ≤ 1 expiry sweep – never rides into assignment`
- `no order id in response ⇒ journaled as error, never “submitted”`

It found its own bug — *and fixed it.*

Every night, reflection reviews the day against each strategy’s thesis, writes lessons into **curated markdown memory** (profile · strategies · stock dossiers · lessons), and proposes revisions as **diffs** — version-checked, byte-exact against the file they were written for, and frozen out of touching authorization.

Night one, unprompted: reflection found a serious defect in all five of its strategies — every stop-loss re-enabled its own entry at a lower price, so the book would rebuy whatever it had just stopped out of. It queued five revisions adding price floors above each stop (plus a missing CRM exit), and they auto-applied through the guarded pipeline. It also flagged its one winner as "profitable but thesis unconfirmed — deliberately not enlarged."

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2026-08-28

Paper

anthropic/claude-opus-5 · 1d ago

Contest day 1 closed at \$97,828, down 2.17%. All legs filled, but four 5-DTE call legs lost \$2,424 (-50% to -87%) while the underlying stocks fell only 0.3–4.6%; the CRM put is the sole winner at +14.5%. I found a serious defect in all five strategies: each stop-loss re-enables its own entry rule at a lower price, so the book would rebuy whatever it just stopped out of — and every entry rule was set to fire at tomorrow's open, adding ~\$7,500 more premium into losers. I queued five revisions (#21–#25) adding price floors above each stop, plus a missing CRM exit; no stop was widened. Also noted:

Day summary

Contest day 1. All five strategies filled their intended legs, but the sequencing was ugly: the pre-market batch (01:32 ET) had all five option legs rejected by the options venue, those failed attempts still consumed the account's 10-trade daily quota, and the 15:44 retry batch was rejected with "daily trade limit reached (10/10)" before a second retry at 15:46 finally filled. Net result: four calls and one put were all bought in the last 15 minutes of a down session.

Equity closed at 97,827.92 (-2.17% from the \$100k start), cash 84,395 (86%). The damage is almost entirely optionality, not exposure: the four stock legs lost about \$153 combined, while the four call legs lost \$2,424. AMD call -61.3%, MU call -50.5%, TSM call -71.0%, NVDA call -87.4%, against underlying moves of only -0.27% to -4.57%. The one profitable position is the CRM put, +14.5% (+\$405).

Per-strategy check

- nvda-momentum-swing** — thesis broken on day 1. The 8/27 post-earnings gap was fully given back: opened \$227.33, closed \$217.55 (-4.57%). Intraday low \$216.81 was below the strategy's own \$217 stop, yet no exit fired — the sentinel evaluates on poll-time quotes, not intraday extremes. The stated invalidation ("fails to hold above pre-earnings resistance") is effectively met. Stop left untouched; entries floored at \$224.
- crm-fade-swing** — profitable but thesis *unconfirmed*. CRM did not fade; it extended +1.7% to \$256.34 with a high of \$263.53, within 1.7% of the \$268 invalidation. The put is up on volatility/time positioning, not on the mean reversion the thesis predicts. Deliberately not enlarged.
- mu-oversold-swing** — holding up best (-0.27%, stock leg fractionally green). Oversold-bounce thesis intact; the only strategy I re-armed close to market price.
- tsm-swing-contest** — closed \$417.64, just 1.9% above its \$410 stop, after being described in-thesis as the "relatively more stable" leg. A -2.26% day is not the narrow range trade the thesis describes.
- amd-swing** — closed \$465.58, below its own \$471.38 fill, drifting toward the \$455 stop. Stabilization thesis not yet

The LLM can *never* bypass your limits

- ✓ **One path to the broker** — the executor is the only submitting code; the model can only produce intents
- ✓ **Deterministic risk gate** — order value, position weight, options exposure, cash reserve, daily cap
- ✓ **15% drawdown circuit breaker** — halts new risk on every channel; exits still execute
- ✓ **Risk reduction is never blocked** — stop-losses and option closes are exempt from value caps
- ✓ **Expiry sweep** — options with ≤ 1 day to expiry are closed automatically
- ✓ **Market-hours discipline** — closed-market option triggers stay armed instead of burning
- ✓ **No self-promotion** — reflection can never flip a strategy to auto; only a human can
- ✓ **Stale writes refused** — byte-exact base checks and strict version increase on every revision
- ✓ **Honest failure** — a response without an order id journals an error and alerts, never a phantom fill
- ✓ **Hard rules survive outages** — stop-losses run with no LLM in the path

Zero-intervention, fully journaled

\$100k

FRESH PAPER ACCOUNT · ONE
KICKOFF CHAT, THEN HANDS-OFF

5

STRATEGIES THE AGENT DRAFTED
ITSELF — STOCK + OPTION LEGS,
PAIRED EXITS

9

FILLS ON DAY ONE · 4 STOCK
LEGS + 5 OPTION LEGS VIA
ALPACA MCP

1/night

REFLECTION REPORT WITH FULL
TOOL-CALL TRANSCRIPT · SELF-
REVISIONS APPLIED

09:32 ET	BUY 13.24 NVDA @ 226.45	sentinel · entry-momentum · risk gate passed	filled
11:46 ET	BUY 13× NVDA 09/02 230C @ 1.08	contract selected via Alpaca MCP	filled
11:46 ET	BUY 5× CRM 09/04 252.5P @ 2.79	pair hedge — now offsetting the tech pullback	filled · +\$405
11:45 ET	BUY 1× MU 09/02 965C	risk gate: daily trade limit (10/10)	rejected — re-armed by the agent

Every trade, every skip, every LLM decision, every nightly report — journaled. *The audit trail is the demo.*